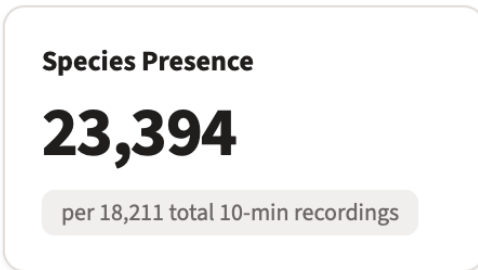


Streamlit App Edits

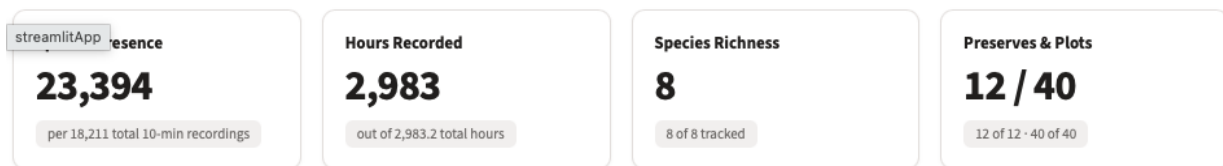
Main Page:

1.)



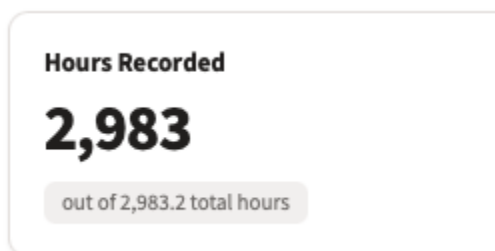
- Change subtext to 'detections from 18,211 10-min recordings'

2.)



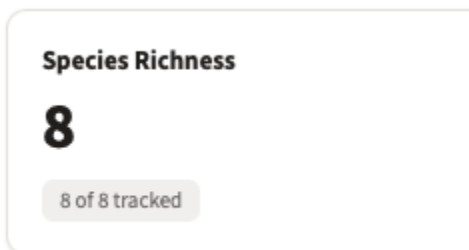
- Increase font of headings in each card, as well as subtext, by 2px

3.)



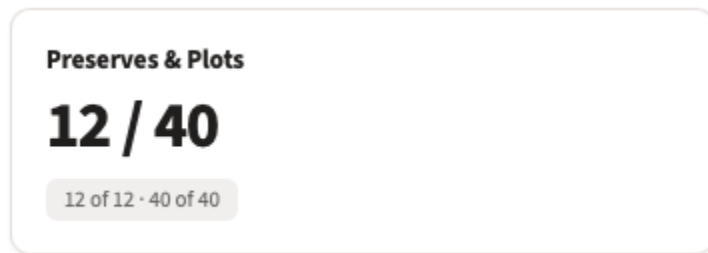
- Round up or down for total hours subtext

4.)



- Subtext should be 'out of 8 tracked' (keep as the total number of species only in the subtext)

5.)



- Instead of '12/40', split the card so that its 'Preserves 12', with subtext 'out of 12', and 'Plots 40', with subtext 'out of 40'. Keep both Preserves and Plots on the same card.

6.)

Total Detections Over Time

PER SEASON

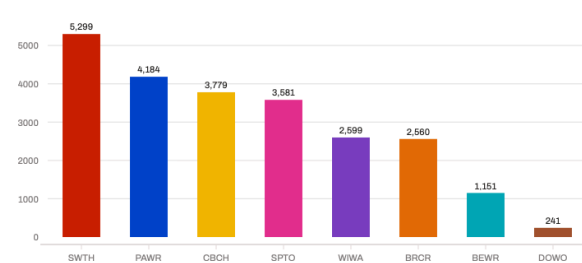
Species detections in 10-min recordings, raw count per season. Not corrected for survey effort.



Total Detections By Species

RANKED

Ranked across the selected range. Not corrected for survey effort.



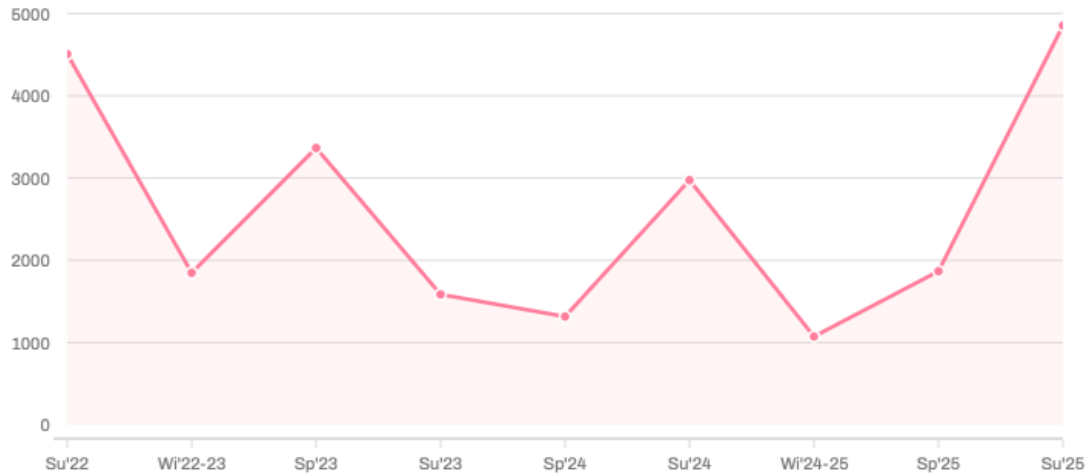
- Move 'PER SEASON' and 'RANKED' to directly under the chart Title. No space needs to be added. Just move it to the far left instead of right.
- Change 'PER SEASON' to 'PER SEASON OVER TIME'
- Change 'RANKED' to 'AGGREGATED & RANKED'

7.)

Total Detections Over Time

PER SEASON

Species detections in 10-min recordings, raw count per season. Not corrected for survey effort.



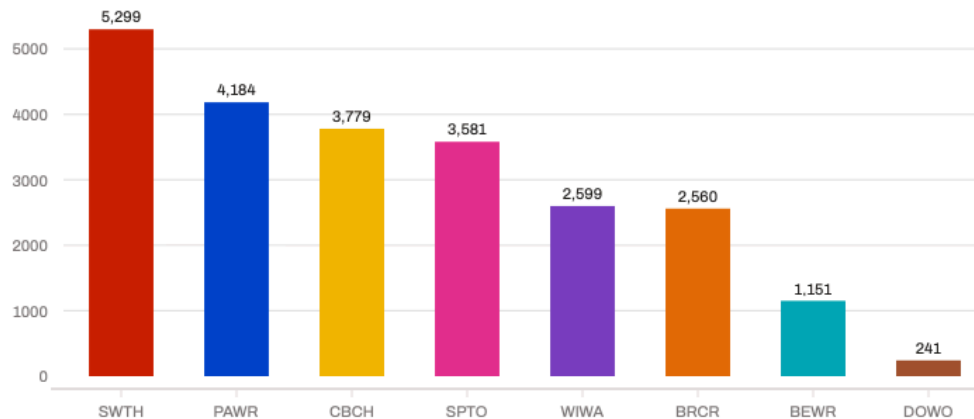
- Add x and y axis labels/titles. X-axis = 'Season-Year', y-axis = 'Count'
- Change the subtext from 'Species detections in 10-min recordings, raw count per season. Not corrected for survey effort.' to 'Total raw count of species detections per season, over the years. NOT corrected for variability in survey effort (days recorded & # of devices)'
- Make the dots on the graph 1px larger, so they're visible when only one year is selected
- Also, add the filter parameter info to the hover over box. Like: Winter 2022-2023, Conf 0.3, All sites, all treatments. needs to be concise/compact, and would be great to write out the full season work and year(s)

8.)

Total Detections By Species

RANKED

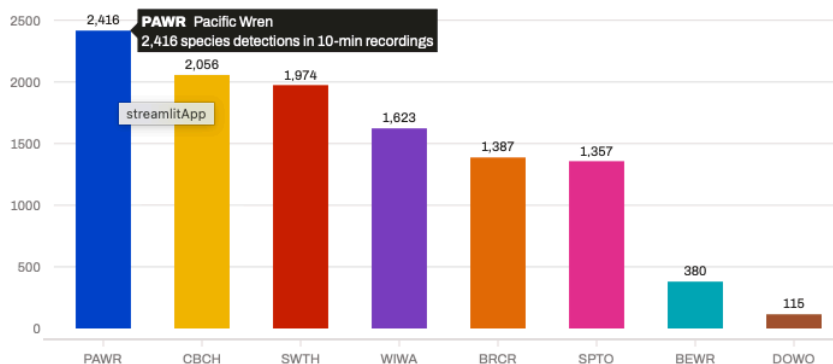
Ranked across the selected range. Not corrected for survey effort.



Total Detections By Species

RANKED

Ranked across the selected range. Not corrected for survey effort.



- Add x and y axis labels/titles. X-axis = 'Species Code', y-axis = 'Count'
- Change 'Ranked across the selected range. Not corrected for survey effort.' to 'Aggregated & ranked total raw counts per species. NOT corrected for variability in survey effort (days recorded & # of devices)'
- When someone hovers their mouse over the bar and the species info appears, add the scientific name after the common name. For example, 'PAWR: Pacific Wren, *Troglodytes pacificus*'. Add the colon after the species code. Also, add the filter parameter info to the hover over box. Like: 2022-2023, All seasons, Conf 0.3, All sites, all treatments. But needs to be concise/compact

9.)

SPECIES

streamlitApp

8/8

CLEAR

BEWR

BRCR

CBCH

DOWO

PAWR

SPTO

SWTH

WIWA

- When someone hovers their mouse over the species code button, have the full species common name and scientific name appear. For example, 'PAWR: Pacific Wren, *Troglodytes pacificus*'

10.)

Daily Occupancy (%)

Daily % Hourly % Count

Share of sampling days a species was detected

PRE-TREATMENT TYPE none POST-TREATMENT TYPE patch cut, thinning · patch cut, thinning, snags, cwd +2 more

PLOTS all 40 plots across 12 preserves

SPECIES CODE	SUMMER 2022	WINTER 2022-23	SPRING 2023	SUMMER 2023
SAMPLING DAYS	30	47	14	15
BEWR	90	55	86	80
BRCR	100	89	100	100
CBCH	100	98	100	100
DOWO	87	21	71	40
PAWR	93	96	100	93
SPTO	100	49	100	100
SWTH	93	0	100	100
WIWA	70	19	100	100

0% 100% NA = NOT SURVEYED

Daily occupancy = days detected ÷ sampling days. A detection counts when BirdNET scores the species at confidence ≥ 0.3 anywhere in a recording; one sampling day is a distinct calendar date on which any selected plot recorded, and a species claims that date if it was heard at any selected plot. Both sides are distinct dates, so five recorders running the same sixteen dates is sixteen sampling days, not eighty. Cells are shaded 0–100%. NA means the season had no sampling effort at the selected plots, not that the species was absent.

- Title should be 'Daily Occupancy Rate (%)'
- Move 'plots' to below 'pre-treatment type' and 'post-treatment type', above the line.
- If a treatment type is long enough to include '+2 more' or whatever the number is, we still need to see these. There should be an expand option so the full list can be seen.
- For the subtext - 'Share of sampling days a species was detected' - change to something like 'rate of daily species occupancy over sampling effort' -> is this an accurate claim?
- Make these changes to the text below the graph: 'Daily Occupancy = days a species was detected ÷ total sampling days at given location. (new line) A detection is counted when BirdNET scores the species at confidence ≥ 0.3 anywhere in a 10-min recording. One sampling day is a distinct calendar date on which any selected plot recorded, and a species claims that date if it was heard at any selected plot. Both sides are distinct dates, For example, five recorders running the same sixteen dates is sixteen sampling days, not eighty. Cells are shaded 0–100%. NA means the season had no sampling effort at the selected plots, while '0' means the species was not detected.'
- the highlighted section doesn't make sense. What are you trying to say here?

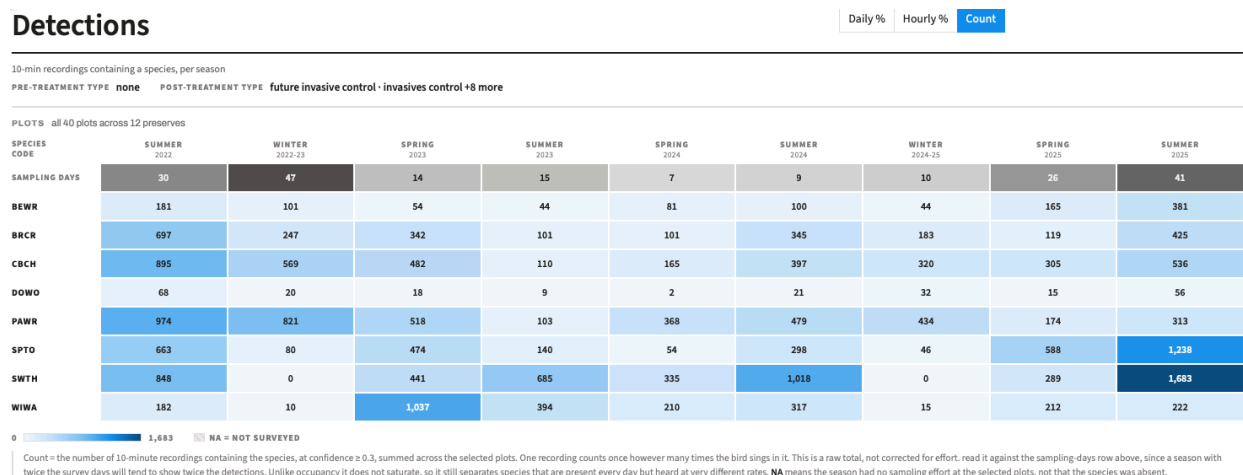
- A user should be able to hover over each cell in the heatmap to get the general stats/parameters for that cell, including species name, general filter parameters used, season/year
- On the top x-axis, season and year, year should be the same font size as season, but not bolded.
- X-axis and y-axis labels should be included

11.) Hourly Occupancy (%)

- Title should be 'Hourly Occupancy Rate (%)'
- Same changes as Daily %, except the text sections will be different
- Change 'Share of recorded hours a species was detected' to something more descriptive and accurate. Still 1-2 sentences
- Change bottom text to: '*Hourly Occupancy = hours detected ÷ sampling hours. A detection is counted when BirdNET scores the species at confidence ≥ 0.3 anywhere in a 10-min recording. One *sampling hour* is a distinct (date, clock-hour) on which any selected plot recorded, and a species claims that date if it was heard at any selected plot. Both sides are distinct dates, so five recorders running the same sixteen dates is sixteen sampling days, not eighty.* Cells are shaded 0–100%. NA means the season had no sampling effort at the selected plots, while '0' means the species was not detected.'
- Is this per day, or at certain times of the day, or how often species are detected within an hour? - this is all unclear. '*Hourly Occupancy = hours detected ÷ sampling hours*' needs to be more specific/descriptive.
- Is the yellow highlighted part even relevant for this plot?

12.)

Detections

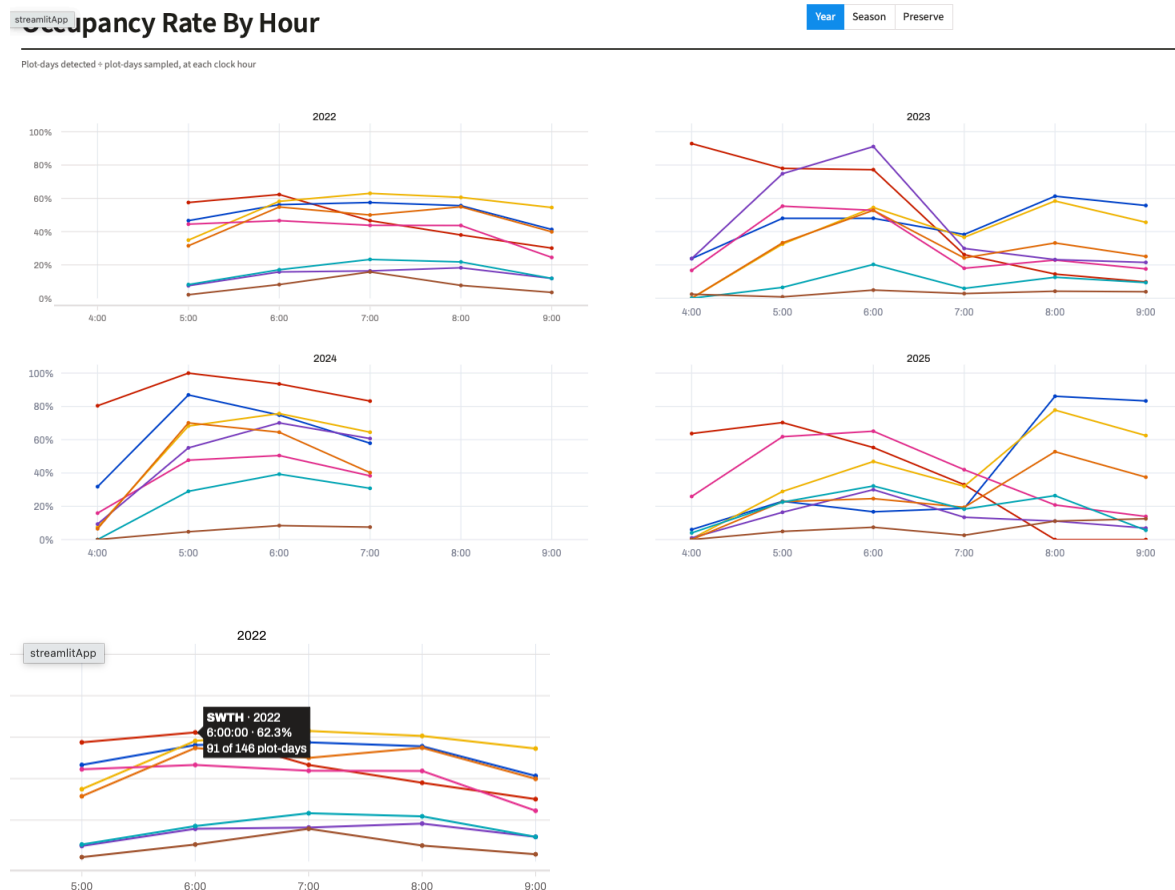


- 'Detections' is vague. Maybe 'Total Detections'
- Make these changes: 'Number of 10-min recordings containing a species was detected at least once, per season-year.'
- Make these changes: '*Count = the number of 10-minute recordings containing the species at least once, at confidence ≥ 0.3 , summed across the selected plots. A species detection only counts once in a 10-min recording, no matter how many times the species vocalizes within that time frame. This is a raw total, not corrected for survey effort (review grey sampling days), nor is it corrected for species vocalization rates (some*

species vocalize more than others). NA means the season had no sampling effort at the selected plots, while '0' means the species was not detected.'

13.) Increase the size of the subtext for all plots

14.)

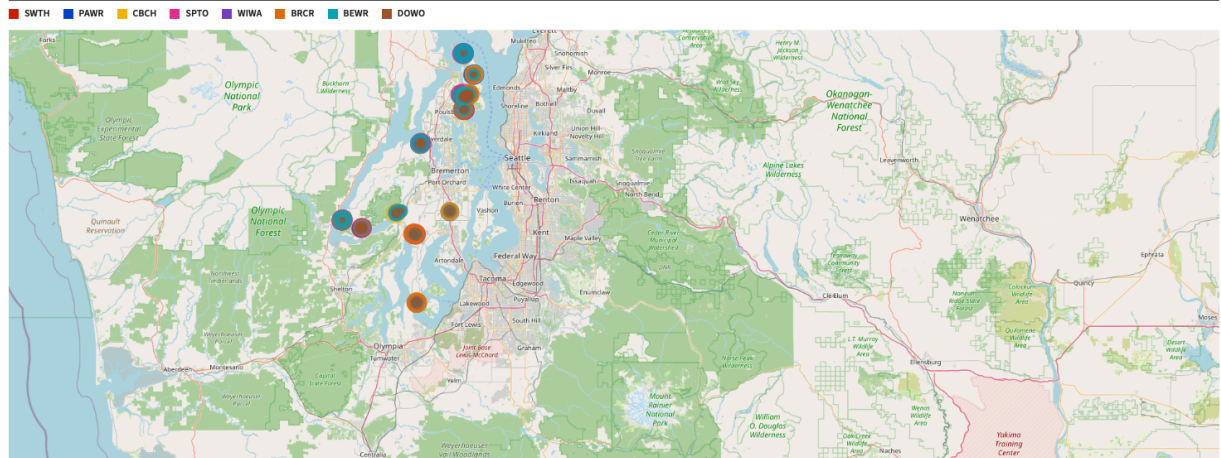


- Title should be 'Occupancy Rate By Hour (%)'
- 'Plot-days detected ÷ plot-days sampled, at each clock hour' - We need a summary that makes more sense to new, non-technical audiences.
- The x-axis and y-axis need labels for each plot. X-axis = 'Hour', y-axis = '% Occupancy'
- There should be a short explanation of metrics at the bottom of this graph, like we have with the heatmaps
- At the top, before the graphs and below the short description, there should be a species legend, like at the top of the 'Species Presence By Plot' map.
- User should be able to hover over any part of the line to get the stats (not just the dot).
- The hover box should include the full species name.
- Remove some white/empty spaces between the graphs, when going from left to right. Not too much.
- Make the font size for the titles 2px bigger
- Make all these changes to the 'Season' and 'Preserve' plots too.

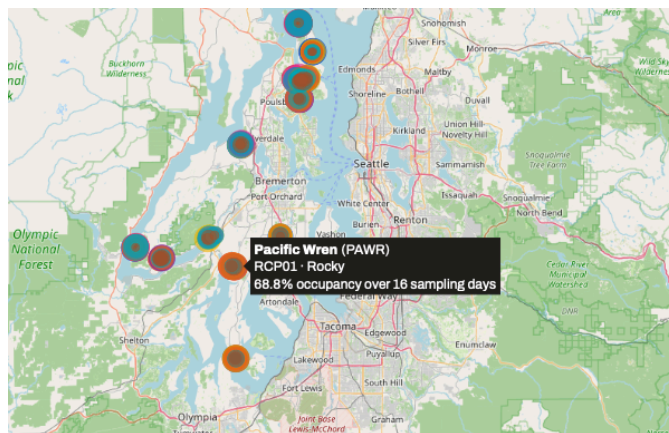
15.)

Species Presence By Plot

OCCUPANCY PER SPECIES



40 plots across 12 preserves. Bubble size is occupancy over the selected seasons.



- Move 'Occupancy Per Species' below the title, left justified instead of right.
- For the hover, make it the same style as all the other hover boxes, ex: 'PAWR: Pacific Wren, *Troglodytes pacificus*', plus the other stats that are already there, plus dates.
- '40 plots across 12 preserves. Bubble size is occupancy over the selected seasons.' -> add a short sentence about occupancy rate, and briefly describe what this map does (shows the range of recording devices and species occupancy at each site).

16.)

287,399 raw detections across 18,211 recordings · confidence thresholds computed per-row from the source table · detection metric is summed 0/1 presence per recording × species.

> Methodology: what each control and measure means

- Remove this -> '287,399 raw detections across 18,211 recordings · confidence thresholds computed per-row from the source table · detection metric is summed 0/1 presence per recording × species.' BUT, we should include an explanation of the detection metric in the methodology section -> 'Detection vs No Detection Metric: a species is counted as 'detected' if it's equal to or greater than the selected confidence threshold. It's considered 'not detected' if it's below that selected interval. A species may

be truly present but counted as 'not detected', depending on confidence threshold settings'

- Make the font of the Methodology section 2px bigger
- In Methodology, capitalize all definition headings. So 'Raw detections' becomes 'Raw Detections'. Do not bold case the definition headers. Only the section headers.
- Make the following changes:
 - Species Presence - 'At least one species detection in one 10-min recording, counted 0/1, depending on confidence threshold setting. A bird heard forty times in a ten-minute file counts once; a file holding three species contributes three.'
 - Raw Detections -> Change to 'Raw 3-sec BirdNET-Analyzer Detections'
 - Add 'Raw 10-min Detections', because we use 10-min detections in most of our graphs, not 3-sec.
 - Sampling Days - 'Distinct dates where recording occurred at a selected plot. The union across plots, not the sum: five recorders over the same sixteen dates is sixteen days.'
 - Days Detected - 'Distinct dates the species was detected at a selected plot.'
 - Daily Occupancy Rate (%) - 'Days detected ÷ sampling days. Example: Grovers in Summer 2022: BEWR on 14 of 16 dates, so 87.5%.'
 - Hourly occupancy (%) - PLEASE IMPROVE based on the above edited language.
 - 'It rises as you add plots - One plot hearing a species claims the day for all of them. Pooled over 40 plots most species read 100%, so compare one plot or preserve against itself.' -> REMOVE
 - 'NA vs 0% - 0% means no species detections at that confidence threshold. NA means no sampling effort (recordings) during that time.'
 - 'What it does' -> change to 'Confidence Threshold - Keeps only detections scoring at or above the chosen value. A higher threshold keeps only high-confidence detections, which tends to reduce false positives (raising precision) but also discards some true positives that scored low (lowering recall).'
 - Change 'HEADER FIGURES: EFFORT AND COVERAGE' to just 'EFFORT AND COVERAGE'
 - Number of Recordings, Hours Recorded, Time of Day, Species Richness -> keep same, just capitalize heading
 - 'What the effort cards respond to - Recordings and Hours recorded follow the plot, season, year and treatment filters, but not species or confidence: a recorder ran for the same time whichever species you ask about.' -> REMOVE entire text row
 - Preserves & Plots -> the current definition is wrong. It's simply the plots and preserves that are being filtered and looked out. Fix please
 - TREATMENT GROUP AND TREATMENT TYPE -> redo this entire section. It doesn't make sense. Have a definition for the Treatment Groups - control, pre-treatment, and post-treatment, and what these mean / and a definition for 'Treatment Types - the type of work being done at the different treatment plots, where applicable.'

- For 'Reading The Charts' -> Redo this entire section. name each chart as the left column header and then explain it, using the language styles above. There should be 9 definitions for 9 plots.

17.)

******'Daily Occupancy' makes it sound like it's the occupancy rate on a certain day. But we're actually looking at a season-year - the occupancy rate over all the days in that season-year. Is there a better descriptive title we could use? Same with 'Hourly Occupancy' - is there a better title?

Filter Panel:

1.) For the 'Season' buttons, add 'Autumn' and grey it out until we have autumn data, like is done in the design document. The buttons could be 2 rows of 2.

2.) Capitalize each header, for example, 'Year range' becomes 'Year Range', and 'Confidence threshold' becomes 'Confidence Threshold', etc.

3.)

Detection unit

☐ Species Presence ☒ Raw Detections

- Change 'Raw Detections' to 'Raw Detections (3-sec BirdNET-Analyzer)', and 'Species Presence' to 'Species Presence (10-min recordings)'

Testing

1.)

LUC BIOACOUSTICS

Filters

TIME

Year range **2022 – 2025**

2022 2023 2024 2025

Season **3/3**

CLEAR

Spring Summer Winter

DETECTION QUALITY

Confidence threshold **≥ 0.30**

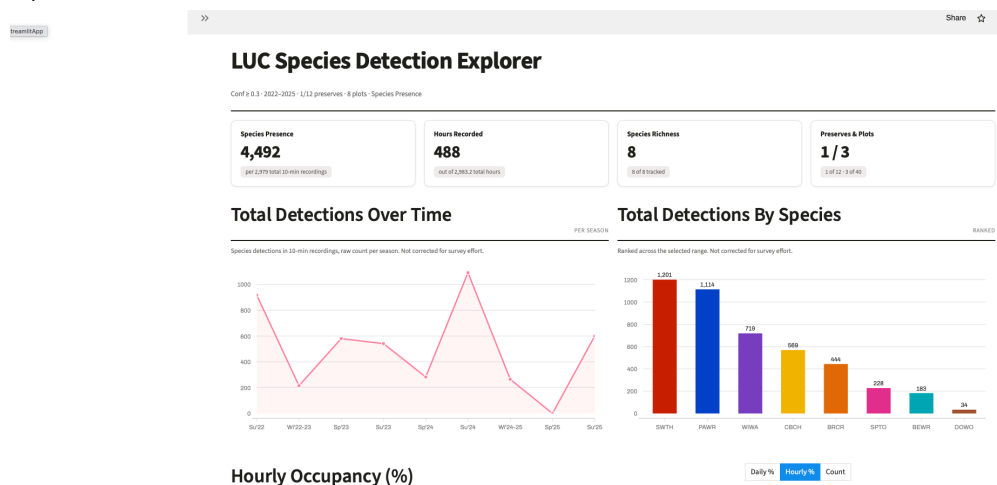
0.3 0.5 0.7 0.9

Detection unit

☒ Species Presence ☐ Raw Detections

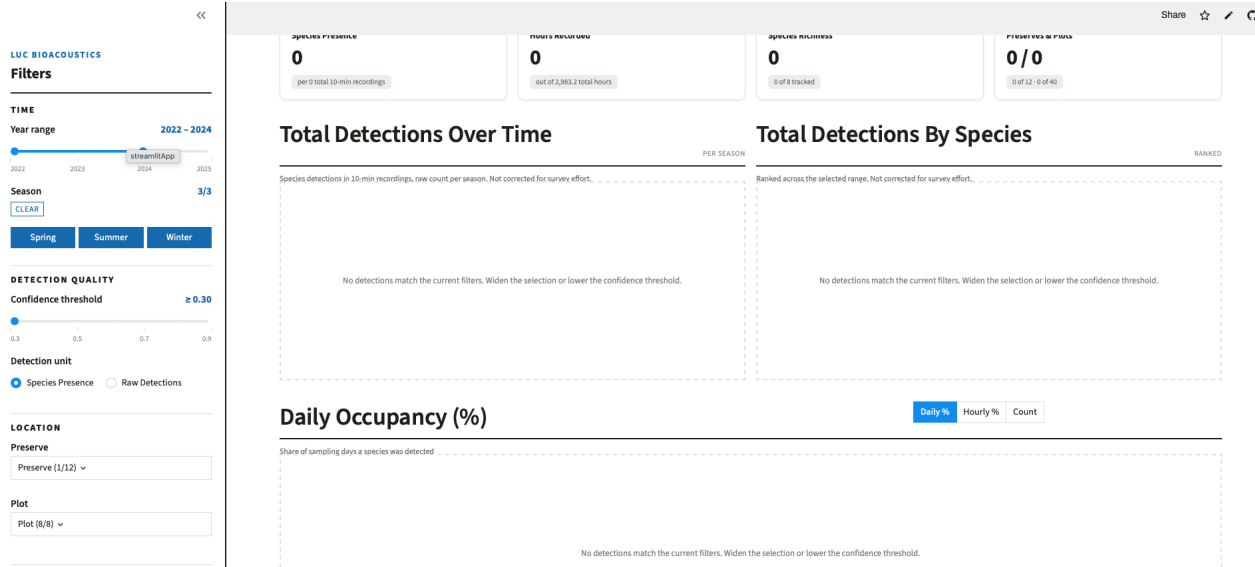
- When I move the bar to expand or shrink the filter panel, it sometimes highlights the text for the entire panel.

2.)

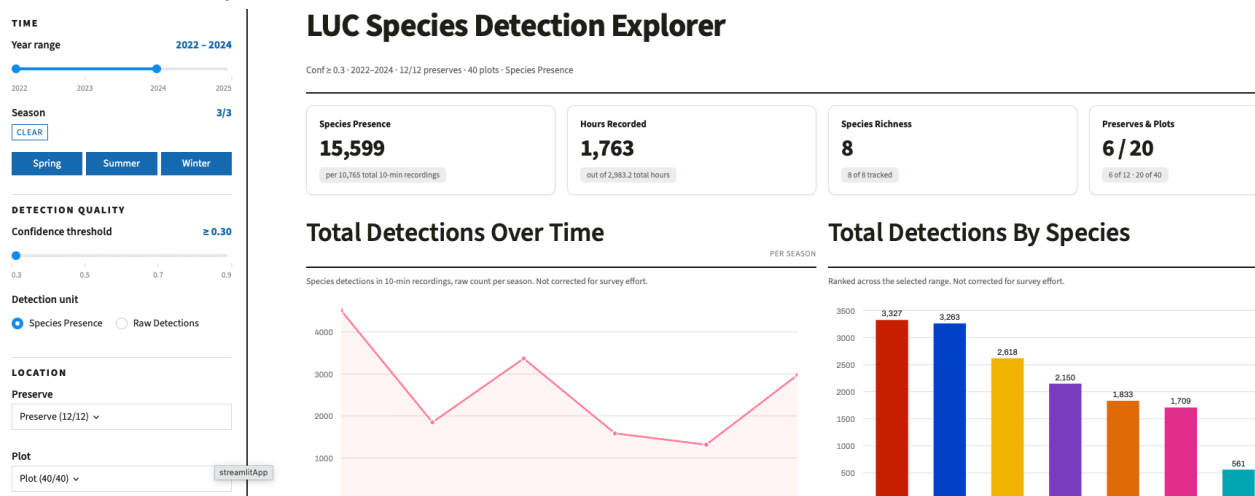


- When i collapse the filter panel, the main screen doesn't take up the whole page, but it should.

3.)

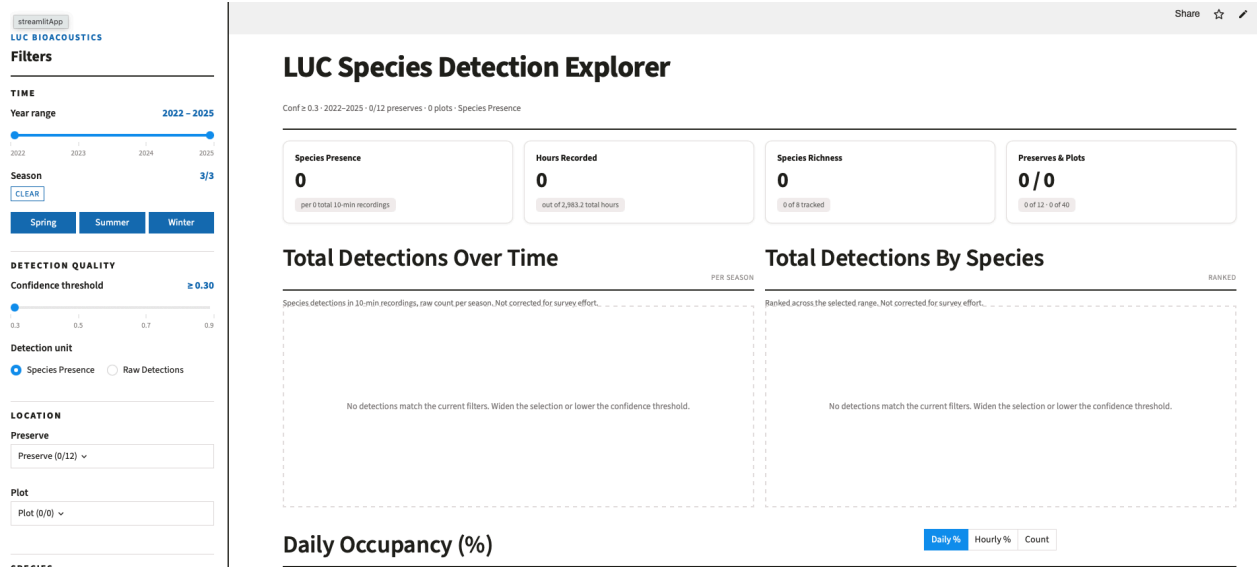


- When I slide the 'Year Range' slider bar, it often breaks the graphs
- The graphs don't always update every time the slider reaches a new date.
- If I slide to just 2022, it doesn't easily slide back to a new date. I have to try more than once. Again, sliding from 2023 to 2024, I have to try it twice before it works. Then I went all the way to 2025, and wanted to move it back to 2024, and it took two tries.



- This is interesting. Here the Year Range is 2022-2024. The Preserves & Plots top panel updated to 6/20. It's also 6/20 for 2022-2023. I assume this is because not all preserves were added yet. We should update the 'Preserve' and 'Plot' 'Location' filter also. The plots/preserves that are not relevant depending on our filter selections should grey out, and the preserve and plot count should also update. So basically, if our Preserves & Plots card at the top updates, our Preserve and Plot filters should also update.

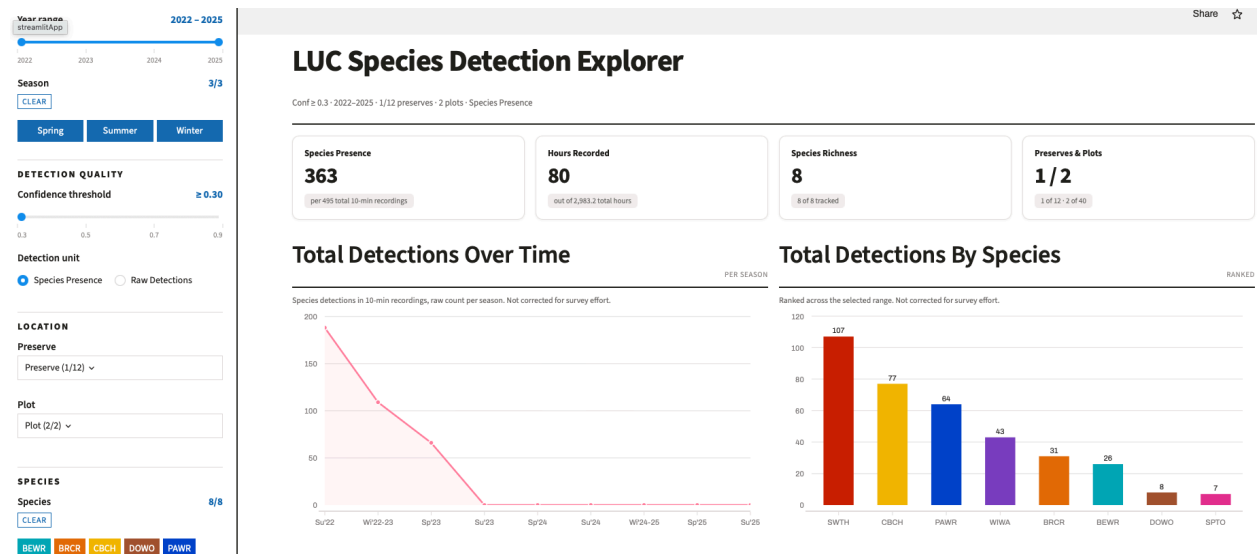
4.)



- I clicked the 'Clear' button for the Season filter and all graphs went empty on the main screen, as expected. I then clicked 'Select All' to see all seasons, and then the main screen did not populate with any data. I had to refresh and start over.

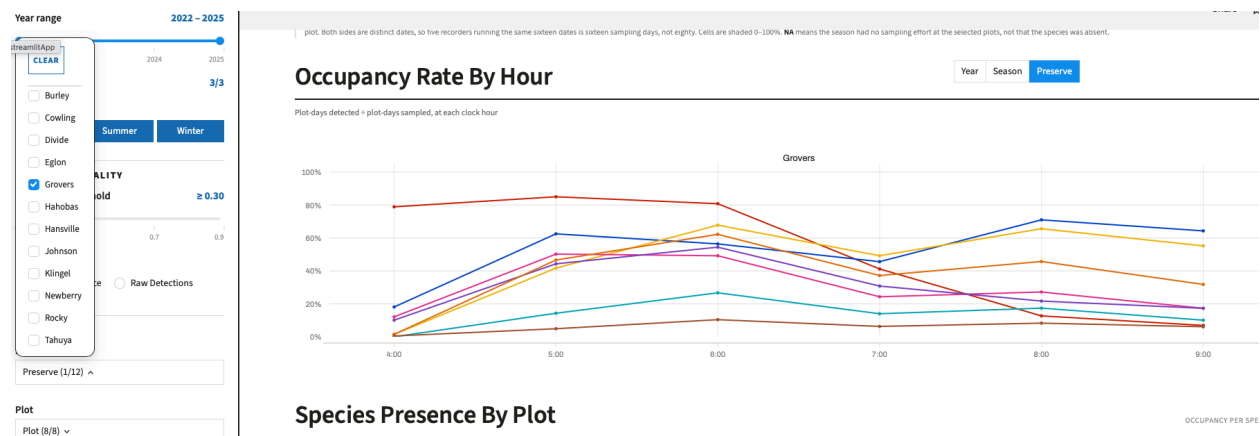
5.) Change 'Clear' button to 'Clear All'

6.)



- I selected the preserve 'Burley' only. The top Hours Recorded card still shows the total 2,983 hours. Do you think we should have the total hours number update to total Burley hours, or keep as the total overall?

7.)



Plot (8/8) ▾

streamlitApp

SPECIES

Species 8/8

CLEAR

BEWR BRCR CBCH DOWO PAWR

SPTD SWTH WIWA

TREATMENTS

Treatment group

Group (3/3) ▾

CLEAR

☒ control

☒ pretreat

☒ posttreat

Reset all filters



- Add a button for Occupancy Rate by Hour line graphs, for 'Treatment Group' and 'Treatment Activity'. There'd be a different hour line graph per treatment group, - so grouped by treatment group type, per preserve. And then grouped by treatment activity, per preserve. Does that make sense?

10.) For the Map

- If I'm zoomed in on a certain location on the map, and then change some of the filters, it should update but stay zoomed in. currently it updates and zooms out.